Cell membranes and transport – AS 9700 Biology Nov 2022

1. Nov/2022/Paper_11/No.16

Which statement is correct for facilitated diffusion **and** active transport?

Protein channels transport molecules across cell membrane

A As the direction of the concentration gradient changes so does the direction of movement of the molecules.

B Molecules always move at the same rate as simple diffusion.

C Specific molecules are transported across a membrane.

D The molecule ATP is required to move specific molecules quickly through proteins in the membrane.

active transport moves materials up their concentration gradient. This process depends on energy from respiration

2. Nov/2022/Paper_11/No.15

A cuboidal epithelium cell has a length of 2 μm.

Surface area = 2x2x6 = 24

volume = 2x2x2 = 8

The cell is an approximate cube shape.

What is the most accurate estimate of the surface area : volume ratio of this cuboidal epithelium cell?

24:8 = 3:1

A 1 : 2 **B** 1 : 3 **C** 2 : 1 **D** 3 : 1

3. Nov/2022/Paper_11/No.17

Equal sized potato pieces were placed into a test-tube and covered with a sucrose solution. The test-tube was left for 30 minutes. All other variables were standardised.

After 30 minutes, the potato piece had not changed in size.

What can be concluded from this result?

water potential

A The concentration of sucrose is the same in the potato and in the solution and there is no more movement of water into or out of the potato. – no net movement of H2O

B The concentration of sucrose is the same in the potato and in the solution and there is no net movement of water into the potato. – water potential is the same

C The water potential is the same in the potato and in the sucrose solution and there is no more movement of water into or out of the potato. – there is no net movement of H2O

D The water potential is the same in the potato and in the sucrose solution and there is no net movement of water into or out of the potato. – the sucrose solution was isotonic

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4. Nov/2022/Paper_12/No.10

Which statements about triglycerides **and** phospholipids are correct?

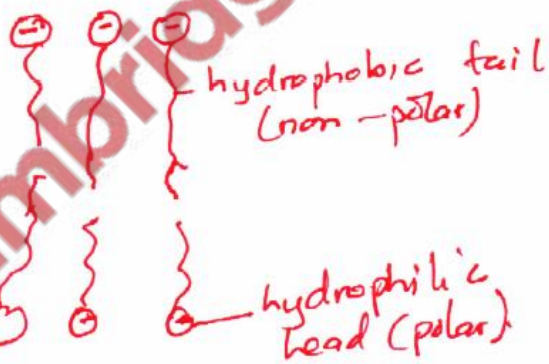
1 Fatty acids in a triglyceride may be saturated or unsaturated but in a phospholipid they are always saturated. – no fatty acids in phospholipids

2 Triglycerides and phospholipids both have a hydrophobic region. – (non-polar)

3 Triglycerides are non-polar molecules and phospholipids are polar molecules.

A 1 and 2 **B** 1 only **C** 2 and 3 **D** 3 only

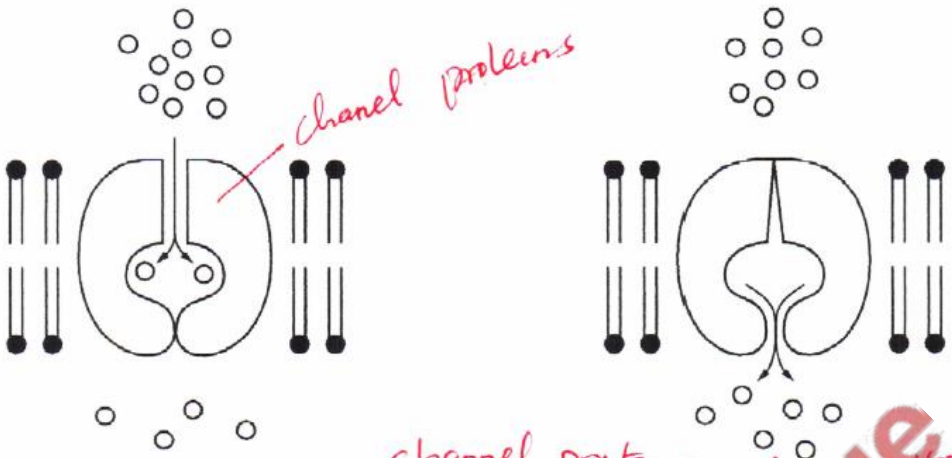
- phospholipids in cell membranes have hydrophobic and hydrophilic ends
- The triglycerides in lipids have a polar and a non-polar end (hydrophobic)



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5. Nov/2022/Paper_12/No.16

The diagram represents a process by which molecules move out of a cell through a cell surface membrane.



channel proteins change their shape to allow molecules to diffuse into the cell

Which process does this represent?

A exocytosis – false

B diffusion – movement of materials down their concentration

C facilitated diffusion – use of chanell proteins to move materials

D osmosis – movement of water into cell from region of high water potential

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6. Nov/2022/Paper_12/No.15

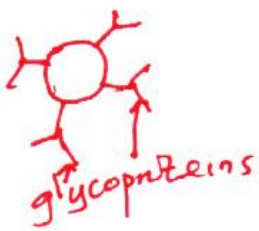
Which functions are performed by glycoproteins on the surface of a cell surface membrane?

1 to act as enzymes catalysing reactions in the membrane – glycoproteins are not enzymes

2 to have a specific site where chemicals can bind

3 to secrete specific chemicals used for cell signalling – APC

A 1, 2 and 3 B 2 and 3 only C 2 only D 3 only



7. Nov/2022/Paper_13/No.15

Which features are correct for active transport **and** facilitated diffusion?

1 The movement of molecules and ions depends on ATP. - facillitated diffusion depend on membrane proteins and not ATP

✓2 They are specific for one type of molecule or ion.

✓3 They use membrane proteins. - faccilitated diffusion

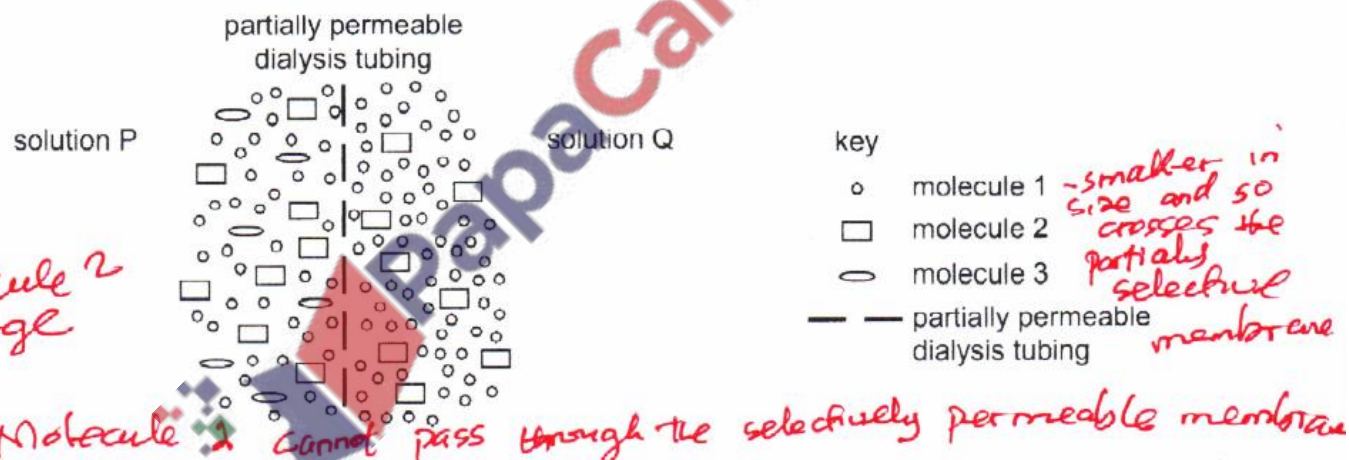
A 1 and 2 **B** 1 and 3 **C** 2 and 3 **D** 3 only

Active transport use ATP.
faccilitated diffusion does not use energy.

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8. Nov/2022/Paper_13/No.16

The diagram represents two solutions, P and Q, that were separated by a partially permeable dialysis tubing.



molecule 2 is large

molecule 1 - smaller in size and so crosses the partially selective membrane

Molecule 3 cannot pass through the selectively permeable membrane

What will be the initial movement of the molecules, 1, 2 and 3, between solution P and solution Q?

	net movement from Q to P	net movement from P to Q	no net movement
A	molecule 1	molecule 2	molecule 3
B	molecule 1	molecule 3	molecule 2
C	molecule 2	molecule 3	molecule 1
D	molecule 3	molecule 1	molecule 2

9. Nov/2022/Paper_13/No.17

A student weighed a cylinder of potato and then put it into a test-tube containing a salt solution.

The potato cylinder was removed from the salt solution after one hour. It was blotted dry and then reweighed. The student recorded that the potato had lost mass.

Which row shows the correct explanation for the results the student collected?

Potato cells in salt concentration loose water by osmosis. The potato cells have a high water potential than salt solution. - So water move out of potato cells. This makes potato cells to be plasmolysed

	water potential of the potato cells before soaking compared to the water potential after soaking	condition of the potato cells after soaking
A	higher	plasmolysed
B	higher	turgid
C	lower	plasmolysed
D	lower	turgid

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